# LECTURE 9
# LINEAR PROCESSES I: WOLD DECOMPOSITION

In this lecture, we focus on covariance stationary processes.

**Definition 1** *(White noise) A process $\{\varepsilon_t\}$ is called white noise (WN) if* $\mathrm{E}[\varepsilon_t] = 0$*,* $\mathrm{E}\left[\varepsilon_t^2\right] = \sigma^2 < \infty$*, and* $\mathrm{E}[\varepsilon_t \varepsilon_{t-j}] = 0$ *for all $t$ and $j \neq 0$.*

**Definition 2** *(Moving average) A process $\{u_t\}$ is called a moving average process of order $q$ (MA($q$)) if*

$$u_t = c_0\varepsilon_t + c_1\varepsilon_{t-1} + \cdots + c_q\varepsilon_{t-q}, \tag{1}$$

*and $\{\varepsilon_t\}$ is a WN.*

A process of the form (1) is called a linear process. The Wold decomposition says that any mean zero covariance stationary process with absolutely summable autocovariances can be represented as the sum of an MA($\infty$) process and a deterministic component.

For a covariance stationary process, we assume that second moments are finite. Let $L_2$ denote the space of random variables with finite second moments. For $X, Y \in L_2$, define the inner product

$$\langle X, Y \rangle = \mathrm{E}[XY].$$

When equipped with such a definition of the inner product, $L_2$ is a Hilbert space. Consider the mean zero covariance stationary process $\{X_t\}$, such that

$$\sum_{j=0}^{\infty} |\gamma(j)| < \infty, \tag{2}$$

where

$$\gamma(j) = \mathrm{E}[X_t X_{t-j}].$$

Define $\mathcal{M}_t$ to be the smallest closed subspace of $L_2$ that contains all elements of the form

$$\sum_{j=0}^{\infty} c_j X_{t-j} \text{ such that } \sum_{j=0}^{\infty} c_j^2 < \infty.$$

The requirement $\sum_{j=0}^{\infty} c_j^2 < \infty$ is to ensure that the elements of $\mathcal{M}_t$ are in $L_2$. Indeed, in this case, the variance of any element of $\mathcal{M}_t$ can be bounded using $\sum_{j=0}^{\infty} c_j^2$ and $\sum_{j=0}^{\infty} |\gamma(j)|$:

$$\begin{aligned}
\operatorname{Var}\left(\sum_{j=0}^{\infty} c_j X_{t-j}\right) &= \sum_{i=0}^{\infty}\sum_{j=0}^{\infty} c_i c_j \gamma(i-j) \\
&\leq 2\sum_{h=0}^{\infty}\sum_{j=0}^{\infty} c_j c_{j+h}\gamma(h) \\
&= 2\sum_{h=0}^{\infty}\gamma(h)\sum_{j=0}^{\infty} c_j c_{j+h} \\
&\leq 2\sum_{h=0}^{\infty}|\gamma(h)|\left|\sum_{j=0}^{\infty} c_j c_{j+h}\right| \\
&\leq 2\sum_{h=0}^{\infty}|\gamma(h)|\left(\sum_{j=0}^{\infty} c_j^2\right)^{1/2}\left(\sum_{j=0}^{\infty} c_{j+h}^2\right)^{1/2} \\
&\leq 2\left(\sum_{j=0}^{\infty} c_j^2\right)\sum_{h=0}^{\infty}|\gamma(h)| \\
&< \infty,
\end{aligned}$$

provided that (2) holds. Furthermore, $\mathcal{M}_t$ is an increasing sequence:

$$\cdots \subset \mathcal{M}_t \subset \mathcal{M}_{t+1} \subset \cdots .$$

Let $P_{\mathcal{M}_t}$ be the orthogonal projection onto $\mathcal{M}_t$. We can write

$$X_t = \widehat{X}_t + \varepsilon_t, \tag{3}$$

where

$$\begin{aligned}
\widehat{X}_t &= P_{\mathcal{M}_{t-1}} X_t, \\
\varepsilon_t &= \left(1 - P_{\mathcal{M}_{t-1}}\right) X_t.
\end{aligned}$$

By the Hilbert space projection theorem, $\widehat{X}_t$ solves the least squares problem. Therefore, $\widehat{X}_t$ can be interpreted as the best one-step-ahead linear predictor of $X_t$ (in the mean squared error sense), and $\varepsilon_t$ is the prediction error. We have $\widehat{X}_t \in \mathcal{M}_{t-1}$, and $\varepsilon_t \in \mathcal{M}_t$, since

$$\varepsilon_t = X_t - \widehat{X}_t.$$

Furthermore, by the orthogonal projection result,

$$\varepsilon_t \in \mathcal{M}_{t-1}^{\perp},$$

where

$$\mathcal{M}_t^{\perp} = \{Y \in L_2 : \langle Y, X\rangle = 0 \text{ for all } X \in \mathcal{M}_t\}.$$

Since $\mathcal{M}_t$ is an increasing sequence, it includes the members of $\mathcal{M}_{t-h}$, and we have that $\varepsilon_t \in \mathcal{M}_{t-h}^{\perp}$ for all $h \geq 1$. Therefore,

$$\operatorname{E}\left[\varepsilon_t \varepsilon_{t-h}\right] = 0 \text{ for all } h \geq 1.$$

Furthermore,

$$\operatorname{E}\left[\varepsilon_t^2\right] = \sigma^2 < \infty$$

and is constant for all $t$ since $\{X_t\}$ is covariance stationary.

Define

$$\mathcal{E}_t = \left\{ \sum_{j=0}^{\infty} c_j \varepsilon_{t-j} : \sum_{j=0}^{\infty} c_j^2 < \infty \right\},$$

where $\mathcal{E}_t$ is a closed linear subspace of $L_2$. Let $P_{\mathcal{E}_t}$ be the orthogonal projection onto $\mathcal{E}_t$. We have

$$X_t = P_{\mathcal{E}_t} X_t + V_t, \tag{4}$$

where, since $\mathcal{E}_t$ is closed and linear,

$$P_{\mathcal{E}_t} X_t = \sum_{j=0}^{\infty} \widehat{c}_j \varepsilon_{t-j} \tag{5}$$

for some sequence $\{\widehat{c}_j\}$, and

$$V_t = (1 - P_{\mathcal{E}_t}) X_t.$$

Since $V_t \in \mathcal{M}_t$, and because of (3),

$$\mathcal{M}_t = \mathcal{M}_{t-1} \oplus \mathcal{S}(\varepsilon_t),$$

where $\mathcal{S}(\varepsilon_t)$ is the linear subspace spanned by $\varepsilon_t$, and $\oplus$ denotes the direct sum:

$$\mathcal{S}_1 \oplus \mathcal{S}_2 = \{x_1 + x_2 : x_1 \in \mathcal{S}_1, x_2 \in \mathcal{S}_2\}.$$

Since $P_{\mathcal{E}_t}$ is the orthogonal projection, $\mathrm{E}[V_t \varepsilon_t] = 0$ by (4) and (5), and therefore $V_t \notin \mathcal{S}(\varepsilon_t)$. Therefore, it must be true that $V_t \in \mathcal{M}_{t-1}$. By the same argument, since

$$\begin{aligned} \mathcal{M}_{t-1} &= \mathcal{M}_{t-2} \oplus \mathcal{S}(\varepsilon_{t-1}), \text{ and} \\ \mathrm{E}[V_t \varepsilon_{t-1}] &= 0, \end{aligned}$$

we deduce that $V_t$ belongs to each of $\mathcal{M}_{t-1}, \mathcal{M}_{t-2}, \ldots$. Let

$$\mathcal{M}_{-\infty} = \bigcap_{t=-\infty}^{\infty} \mathcal{M}_t.$$

We conclude that

$$V_t \in \mathcal{M}_{-\infty} \text{ for all } t.$$

Thus, $V_t$ is an element of any linear subspace $\mathcal{M}_s$, $s \in \mathbb{Z}$, where $\mathbb{Z}$ is the set of integers. The entire process $\{V_t\}$ can be predicted with certainty from an arbitrarily distant past of $\{X_t\}$. Such a process is called deterministic.

We derived the Wold representation for a mean zero covariance stationary process $\{X_t\}$:

$$X_t = \sum_{j=0}^{\infty} c_j \varepsilon_{t-j} + V_t,$$

where $\{\varepsilon_t\}$ is a WN, and $V_t$ is deterministic. When $V_t = 0$ for all $t$, the process $\{X_t\}$ is said to be linearly indeterministic. If $\{X_t\}$ is linearly indeterministic and mean zero, it has the MA($\infty$) representation

$$X_t = \sum_{j=0}^{\infty} c_j \varepsilon_{t-j}.$$

It follows from the projection properties that

$$\begin{aligned} c_j &= \mathrm{E}[X_t \varepsilon_{t-j}] / \sigma^2, \text{ and} \\ c_0 &= 1. \end{aligned}$$

Remarks:

1. The representation is unique with probability one, which follows from the uniqueness of the Hilbert space projection.

2. The WN elements in the Wold decomposition are the one-step-ahead linear prediction errors.

3. Sometimes $\varepsilon_t$'s are interpreted as the fundamental shocks of the economy. Then the impulse responses, $c_j$'s, represent the effect of a shock after $j$ periods. Suppose that the fundamental shocks are given by the unexpected shocks to the agents' information set. Let $\mathcal{F}_t = \sigma(X_t, X_{t-1}, \ldots)$ represent the information of the agents in the economy at time $t$. The predictor $\widehat{X}_t$ is restricted to be a linear predictor and, therefore, is not necessarily equal to $\mathrm{E}\left[X_t \mid \mathcal{F}_{t-1}\right]$. Hence, the true fundamental shocks may differ from the shocks that enter the Wold representation (see, for example, Hansen and Sargent (1991), Chapter 4).